Lab 7: Fuzzing for Input Validation Bugs

Repository: L6_L7_48_daycare_fees (Lab 7) | Module: daycare_fees.py

Step 2: Buggy application output

```
Harsha_R@Harsha UCRT64 ~/L6_L7_48_daycare_fees/Lab 7 (main)
$ py daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 20,25,30
Average Hours: 25

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0

Harsha_R@Harsha UCRT64 ~/L6_L7_48_daycare_fees/Lab 7 (main)
$
```

Figure 1: `python daycare_fees.py` (original code) with valid input (toddler:25, 20,25,30, 3000|250) works as expected.

```
Harsha_R@Harsha UCRT64 ~/L6_L7_48_daycare_fees/Lab 7 (main)
$ py daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler
Traceback (most recent call last):
  File "C:\Users\Harsha_R\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 31, in <module>
    main()
  File "C:\Users\Harsha_R\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 17, in main
    result = parse_enrolment(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Harsha_R\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 2, in parse_enrolment
    age_group, hours = text.split(':')
    ^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)

Harsha_R@Harsha UCRT64 ~/L6_L7_48_daycare_fees/Lab 7 (main)
$
```

Figure 2: The same original code crashes on input without a separator (`toddler`) with an unhandled ValueError.

Step 4: Hypothesis fuzz tests failing on the buggy code

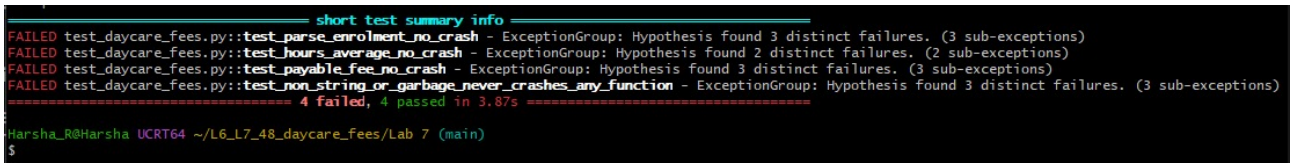


Figure 3: `pytest test_daycare_fees.py` against the original code: 4 failed, 4 passed. Hypothesis found crashing inputs for all three functions.

Step 5: Fixed application output

```
Harsha_R@Harsha UCRT64 ~/L6_L7_48_daycare_fees/Lab 7 (main)
$ py daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 20,25,30
Average Hours: 25

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0

Harsha_R@Harsha UCRT64 ~/L6_L7_48_daycare_fees/Lab 7 (main)
$ py daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler
Age Group and Hours: Invalid input

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 1
Average Hours: 1

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3
Payable Fee: Invalid input

Harsha_R@Harsha UCRT64 ~/L6_L7_48_daycare_fees/Lab 7 (main)
$
```

Figure 4: `python daycare_fees.py` after hardening. Valid input still gives the same results; invalid input now prints "Invalid input" instead of crashing.

Step 6: Fuzz tests passing on the fixed code

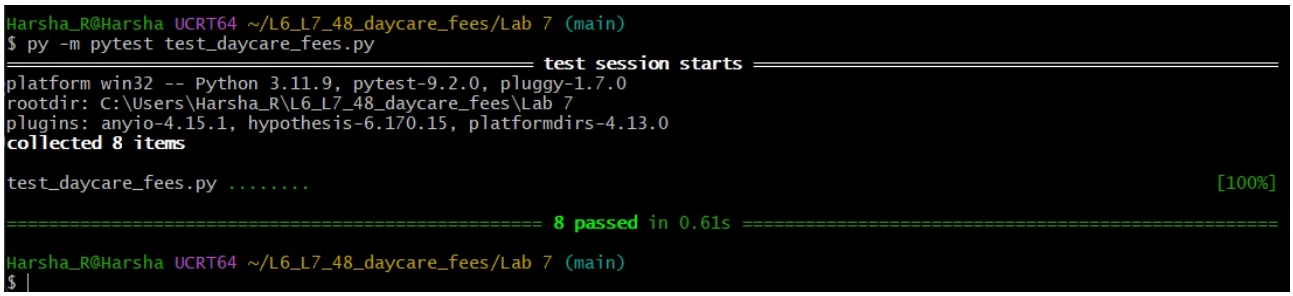


Figure 5: `pytest test_daycare_fees.py` against the fixed code: 8 passed.